LUCAS KIMBER

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TECHNICAL SKILLS

Programming Languages: Python, Bash, Rust, TypeScript/JavaScript, Go, Java, C/C++, SQL
Tools: Linux, Git, GitLab CI/CD, Docker, Kubernetes, Cypress, Prometheus
Methodologies: Test Driven Development (TDD), DevOps, Agile, Object Oriented Programming (OOP), Microservices

EDUCATION

Imperial College London, MSc in Advanced Computing (Merit) Sep 2024 – Nov 2025
Distributed systems, computational optimisation, and software engineering at scale.

Royal Holloway University of London, BSc in Computer Science (First) Sep 2020 – Jun 2024
Advanced algorithms, project and requirements management, and privacy engineering.

WORK EXPERIENCE

Teaching Assistant Sep 2023 – Apr 2024
Royal Holloway University of London *Surrey, UK*
- Supported Python and C programming laboratories for undergraduate students, assisting with debugging and programming fundamentals.

Software Engineer Internship Jun 2023 – Sep 2023
Arqit *London, UK*
- Developed DevOps tooling supporting a post-quantum cloud encryption platform.
- Eliminated 1-2 hours of manual regression testing per PR to the test environment by implementing automated Cypress tests (integrated into GitLab CI/CD pipelines).

Software Engineering Year in Industry Jun 2022 – Jun 2023
ABB Ltd *Cambridgeshire, UK*
- Developed and debugged embedded firmware for industrial transmitter systems in C.
- Proposed and implemented CI improvements by integrating Ceedling unit tests and IAR toolchain metrics into Azure DevOps (using pipelines and Python conversion scripts).
- Achieved the Engineering Development Trust's Platinum Award for the placement.

Freelance GCSE English and Maths Tutor Mar 2021 – Jun 2022
Private *Online*
- Designed and led personalised one-on-one tutoring sessions.
- Improved a student's grades from failing mock exams to all 6's (B equivalents).

PROJECTS

Distributed Runtime targeting FractOS, MSc Dissertation (Python, C++)
- Developed a Ray-like distributed computing runtime for disaggregated data centres.
- Reduced end-to-end latency by 20–80% across benchmark workloads by optimising RPC traffic and scheduling logic across multiple worker nodes.
- Managed multiple Ubuntu servers for development and testing.

Machine Learning Kidney Injury Prediction System, MSc Project (Python, SQL)
- Achieved F3 score of 98%, improving on the NHS algorithm baseline of 70%
- Maintained 99.99% availability and <1s p90 inference latency across a two-week stress test, utilising a Kubernetes deployment and Prometheus monitoring.

Web-Service QoS Optimisation, MSc Research (Python)
- Researched optimising quality of service for web-service composition designs using meta-heuristics.
- Demonstrated optimisation techniques that performed 83% better than baseline methods.
- Developed a web-service workflow framework and used web-scraped data to run experiments.

RSA Encryption Application, BSc Dissertation (Rust, TypeScript)
- Implemented RSA cryptographic primitives from scratch in Rust, and built a TypeScript/Tauri cross-platform interface.